Anuja Raorane

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Affiliation: University of Vienna

Research interests

Planetary science, atmospheric evolution and escape, Venus and Mars, exoplanet habitability, star-planet interactions, atmosphere-surface exchange, planet formation

Education

2025 – Present **PhD in Astrophysics**
University of Vienna – Vienna, Austria
Supervisor: Dr. Kristina Kislyakova.

2018 – 2023 **Master of Science - Bachelor of Science Dual Degree**
Indian Institute of Science Education and Research (IISER),Pune – Pune, India
Specialization in Physics, and Earth and Climate Sciences
GPA: 1.87.

Publications

2024 **Giant planet formation in the Solar System**
A. Raorane, R. Brasser, S. Matsumura, T. C. H. Lau, M. H. Lee, A.Bouvier
Icarus Vol 421, October 2024, 116231.

Research experience

04.2025 – Present **PhD Thesis: Influence of non-thermal escape processes on the evolution of early Earth, Mars, and Venus**
Supervisor: Dr. Kristina Kislyakova, University of Vienna
My research focuses on non-thermal atmospheric escape driven by solar wind and magnetospheric interactions at Earth, Venus, and Mars, processes that play a key role in shaping the long-term evolution of rocky planets. I use a combination of models using numerical tools and validate them against observations.

06.2023 – 08.2024 **Internship: Giant planet formation in the outer solar system**
Supervisor: Dr. Ramon Brasser, CSFK, Budapest, Hungary.
Running N-body simulations based on pebble and planetesimal accretion in the outer solar system gas disk.

11.2023 – 01.2024 **Internship: Spectro-polarimetric data regeneration for Earth**
Supervisor: Mr. Bhavesh Jaiswal, U.R. Rao Satellite Centre, ISRO, Bangalore, India.
Using Planetary Spectrum Generator and MODIS Cloud Data Products to re- produce Earth's spectral features as seen from the SHAPE payload onboard Chandrayaan-3 Orbiter.

06.2022 – 04.2023 **Master Thesis: Formation of Saturn and distribution of its growth times**
Supervisor: Dr. Ramon Brasser, Dr. Shreyas Managave, IISER Pune.
Running N-body simulations in the outer solar system gas disk and taking a data-based approach from pallasite geochronology to put time-constraints on giant planet formation.

Conference Presentations

04.2026 European Geosciences Union (EGU) 2026 General Assembly, Vienna, Austria
Presentation - Non-thermal escape of Venus atmosphere

03.2026 Space Weather Workshop, Graz
Presentation - Non-thermal escape evolution of atmospheres of Earth, Venus and Mars

03.2026 Exploring Exoplanets School, Bangalore, India
Poster - Non-thermal escape of atmospheres on rocky planets

02.2026 Austrian Early Career Conference, Salzburg, Austria
Presentation - Non-thermal escape of atmospheres on rocky planets

09.2025 EPSC-DPS Joint Meeting 2025, Helsinki, Finland
Poster - Giant planet formation in the Solar system

05.2025 Graz-Vienna Exoplanet Scientist Meeting VI, Graz, Austria
Flash Talk - From giant planet formation to rocky planet atmospheres

09.2024 NISER Astronomy Club, Bhubaneshwar, India
Invited Talk (Online) - Formation of giant planets: Insights from Pebble accretion

04.2023 MS Thesis Talk Series, Aakashganga (IISER Pune Astronomy Club), Pune, India
Invited Talk - Understanding Saturn's formation

Honors and scholarships

2023 Best Master's Thesis Award (IISER Pune)
In Earth and Climate Science Department.

2018 – 2023 KVPY Scholarship (Government of India)
A prestigious national fellowship initiated by the Department of Science and Technology to identify and nurture young scientific talent.

Teaching and Supervision

03.2026 – Present — Bachelor's Thesis Supervisor, University of Vienna
Topic: Ion production at Venus and Mars due to non-thermal processes

04.2026 — Basic Qualification in Teaching in STEM
Successfully completed the course on basic qualification in teaching offered by University of Vienna

Technical skills

Programming languages
Proficient in: Python, Fortran, Bash

Software and Tools
LaTeX, Git, Planetary Spectrum Generator (NASA), OpenMP

Languages
English (fluent), German (A2), Hindi (fluent), Marathi (mother-tongue)

Management and Volunteering

04.2026 — Volunteer/ Organizer — Long Night of Research 2026
Kids session organizer — How colorful, large, or wild can a planetary system be when you create it as a bracelet?

12.2025 — Member — Science Organizing Committee
The Wienerwald Astronomy Symposium 2025, ISTA, Austria

10.2025 — Organizer — VISESS Big Picture Event Organising Committee
Seminar and Workshop — When Science Meets Society: Navigating Conflicting Interests

06.2021 – 07.2022 — General Coordinator — DISHA
A student-led initiative to help educate underprivileged kids.

05.2020 – 06.2021 — Coordinator — Navarasa
IISER Pune Dance Club